Rubber_band_analysis

Analysis of Field Data and AI-AI in Rubber-band dynamics

PENPAL Project

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1 1. Data Loading and Preprocessing

1.1 1.1 Load Data

```
## Field data dimensions: 857 15
```

```
## Simulated data dimensions: 30 12
```

```
##
## Field data columns:
```

```
##  [1] "timestamp"            "user"                 "ai"
##  [4] "combined_prompt"      "client_id"            "workshop_id"
##  [7] "language"             "conversation_id"      "edit_distance"
## [10] "user_embedding"       "ai_embedding"         "turn"
## [13] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

```
##
## Simulated data columns:
```

```
##  [1] "turn"                 "user"                 "ai"
##  [4] "client_id"            "workshop_id"          "conversation_id"
##  [7] "timestamp"            "user_embedding"       "ai_embedding"
## [10] "pct_turn"             "user_sentiment_score" "ai_sentiment_score"
```

1.2 1.2 Data Preprocessing

```
##
## === Field Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1             205     857                 4.18            -0.507          -0.593
```

```
##
## === Simulated Data Summary ===
```

```
## # A tibble: 1 x 5
##   n_conversations n_turns mean_turns_per_story mean_user_valence mean_ai_valence
##             <int>   <int>                <dbl>             <dbl>           <dbl>
## 1               6      30                    5            -0.407          -0.296
```

1.3 1.3 Compute Lagged Variables and Cosine

```
## === Lag Variable Verification ===
```

```
## Field data - Turn 1 with non-NA lag (should be 0): 0
```

```
## Simulated data - Turn 1 with non-NA lag (should be 0): 0
```

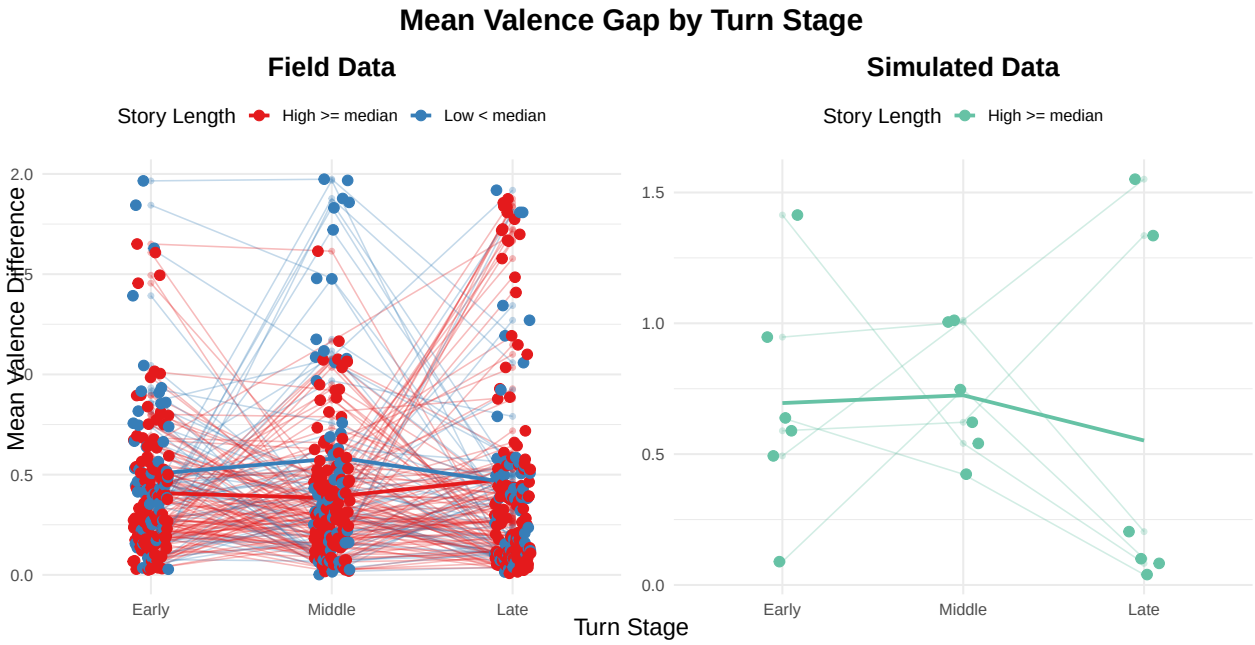
2 2. Temporal Dynamics Analysis

2.1 2.1 Stage-Based Analysis (Early/Middle/Late)

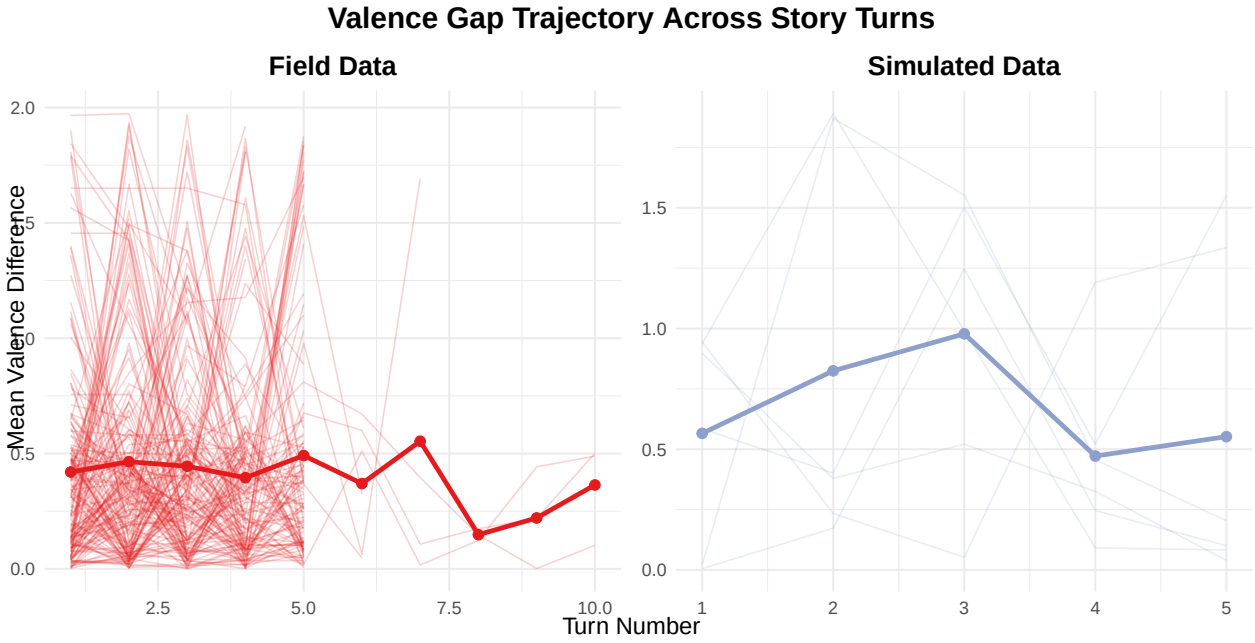
```
## === Mean Valence Gap by Stage ===
```

```
## # A tibble: 6 x 4
##   data_source stage_label mean_gap sd_gap
##   <chr>       <fct>          <dbl>  <dbl>
## 1 Field       Early          0.443     NA
## 2 Field       Middle         0.451     NA
## 3 Field       Late           0.468     NA
## 4 Simulated   Early          0.695     NA
## 5 Simulated   Middle         0.725     NA
## 6 Simulated   Late           0.552     NA
```

2.2 2.2 Stage Valence Gap Plot



2.3 2.3 Turn-by-Turn Valence Gap Trajectory



2.4 2.4 Rubber-Band Dynamics

```
## === Rubber-Band Model: Field Data ===

##
## Call:
## lm(formula = delta_23 ~ delta_12, data = rb_field)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -1.6099 -0.3130 -0.1035  0.1298  1.5559
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.02219    0.04249   0.522    0.602
## delta_12    -0.57784    0.08346  -6.924 8.38e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5604 on 172 degrees of freedom
## Multiple R-squared:  0.218,  Adjusted R-squared:  0.2134
## F-statistic: 47.94 on 1 and 172 DF,  p-value: 8.377e-11
```

```
##
## === Rubber-Band Model: Simulated Data ===
```

2.5 2.5 Turn-by-Turn Rubber-Band Dynamics

```
## === Turn-Level Rubber-Band Model: Field Data ===
```

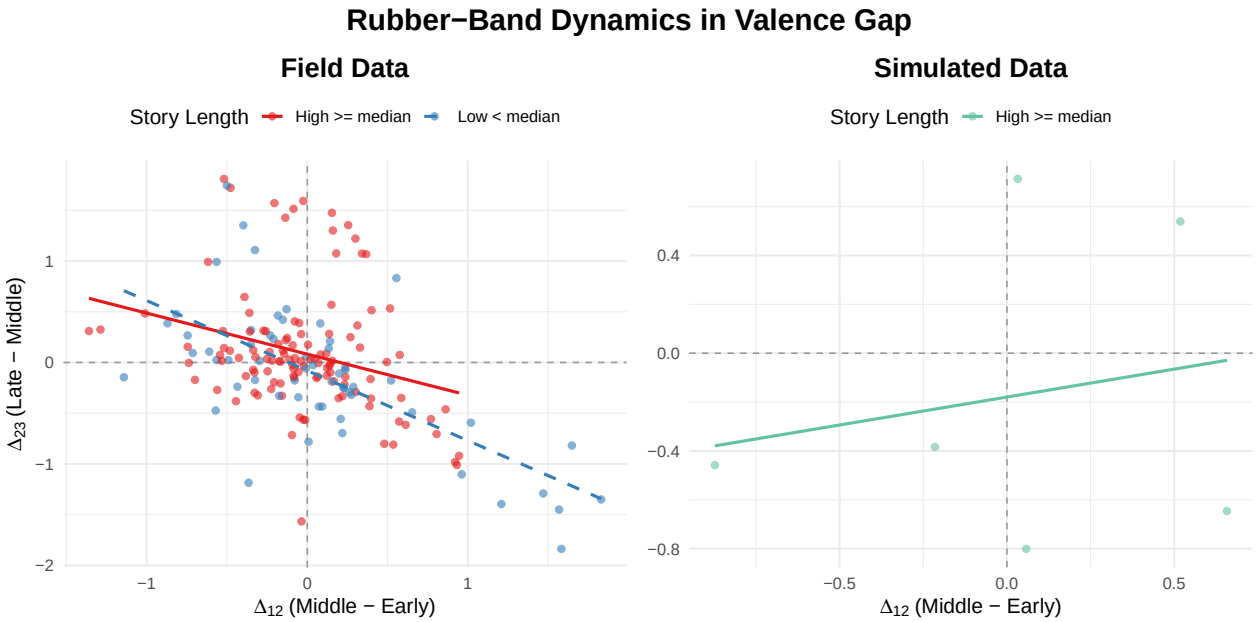
```
##
## Call:
## lm(formula = gap_change_next ~ gap_change, data = rb_turns_field)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.58042 -0.30684 -0.03182  0.22167  1.81851
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.01136    0.02462   0.462    0.645
## gap_change  -0.52836    0.04316 -12.242   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5319 on 465 degrees of freedom
## Multiple R-squared:  0.2437, Adjusted R-squared:  0.2421
## F-statistic: 149.9 on 1 and 465 DF,  p-value: < 2.2e-16
```

```
##
## === Turn-Level Rubber-Band Model: Simulated Data ===
```

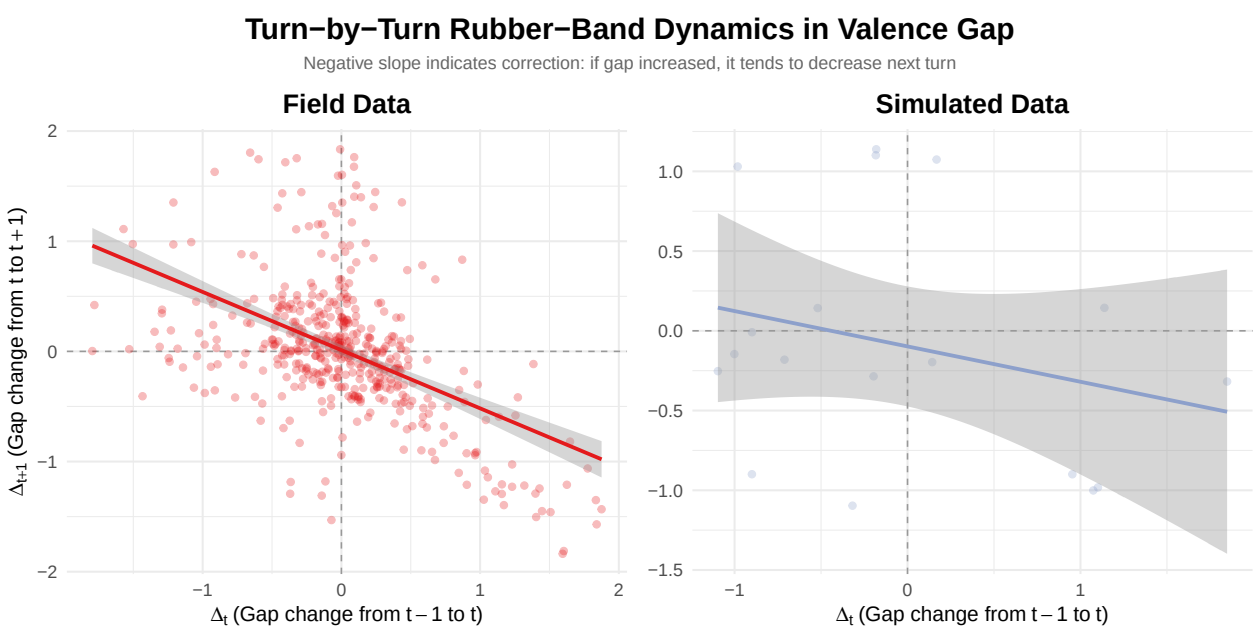
```
##
## Call:
## lm(formula = gap_change_next ~ gap_change, data = rb_turns_sim)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0692 -0.5424 -0.1708  0.4181  1.2092
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.09783    0.17752  -0.551    0.589
## gap_change  -0.22171    0.20273  -1.094    0.290
##
```

```
## Residual standard error: 0.7527 on 16 degrees of freedom
## Multiple R-squared:  0.06955,    Adjusted R-squared:  0.0114
## F-statistic: 1.196 on 1 and 16 DF,  p-value: 0.2903

##
## === Turn-Level Rubber-Band Correlation ===

## Field correlation (gap_change vs gap_change_next): -0.4936882

## Simulated correlation (gap_change vs gap_change_next): -0.2637313
```

2.6 2.6 Rubber-Band Dynamics Plot



2.7 2.7 Turn-Level Rubber-Band Dynamics Plot



3 3. Permutation tests

3.1 3.1 Stage-level Permutation Test

```
## === Stage-Level Permutation Test Results ===
## Observed slope: -0.5778
## Null mean: -0.4956
## Null SD: 0.0827
## Effect size (z): -1
## P-value: 0.163
```

3.2 3.2 Brutal Stage-Level Permutation

```
## === Stage-Level Permutation Test Results - Shuffle Turn-Level Valence Gap first ===
## Observed slope: -0.5778
## Null mean: -0.5577
## Null SD: 0.084
## Effect size (z): -0.24
## P-value: 0.419
```

3.3 3.2 Global Shuffle Stage-Level Test

```
##
## Call:
## lm(formula = d23 ~ d12, data = df_global)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.66487 -0.23794 -0.00151  0.23258  0.79722
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.05638    0.06111  -0.922    0.362
## d12         -0.64577    0.09156  -7.053 2.41e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.379 on 37 degrees of freedom
## Multiple R-squared:  0.5734, Adjusted R-squared:  0.5619
## F-statistic: 49.74 on 1 and 37 DF,  p-value: 2.412e-08
```

3.4 3.3 Turn-Level Permutation Test

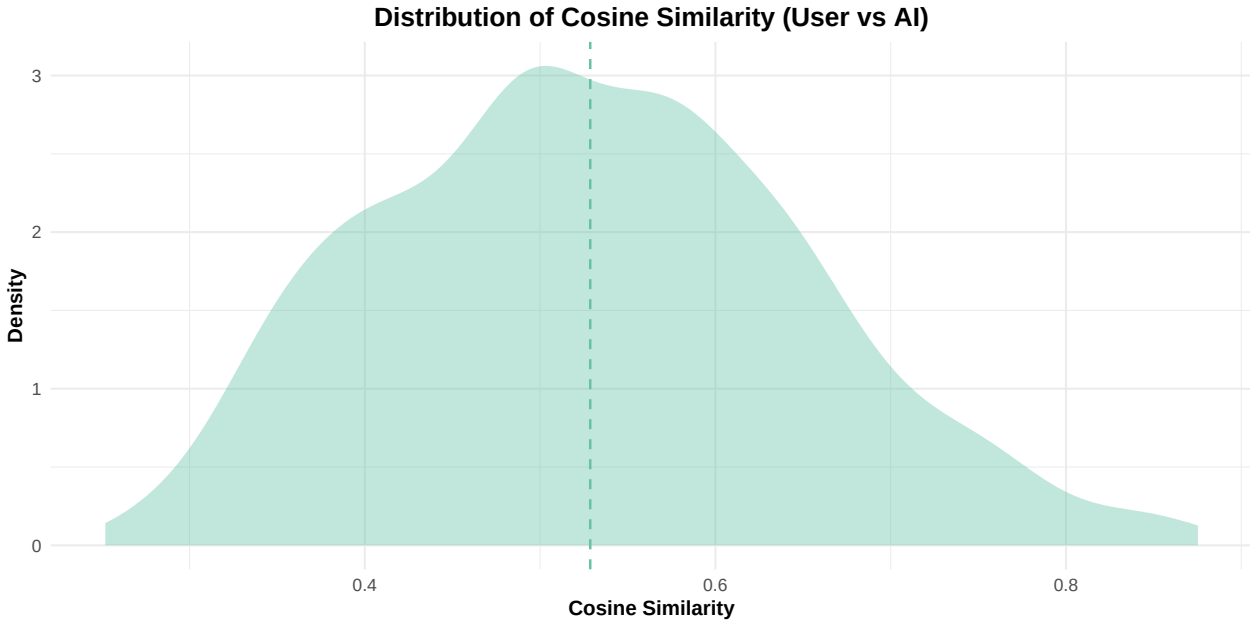
```
## === Turn-Level Permutation Test Results ===
## Observed slope: -0.5284
## Null mean: -0.0125
## Null SD: 0.0745
## Effect size (z): -6.92
```

```
## P-value: <2e-16
```

```
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.440      -0.458   274
## 2 Within-Stage      -0.659      -0.580    19
```

4 4. Cosine similarity alignment

4.1 4.1 Cosine Similarity Distribution

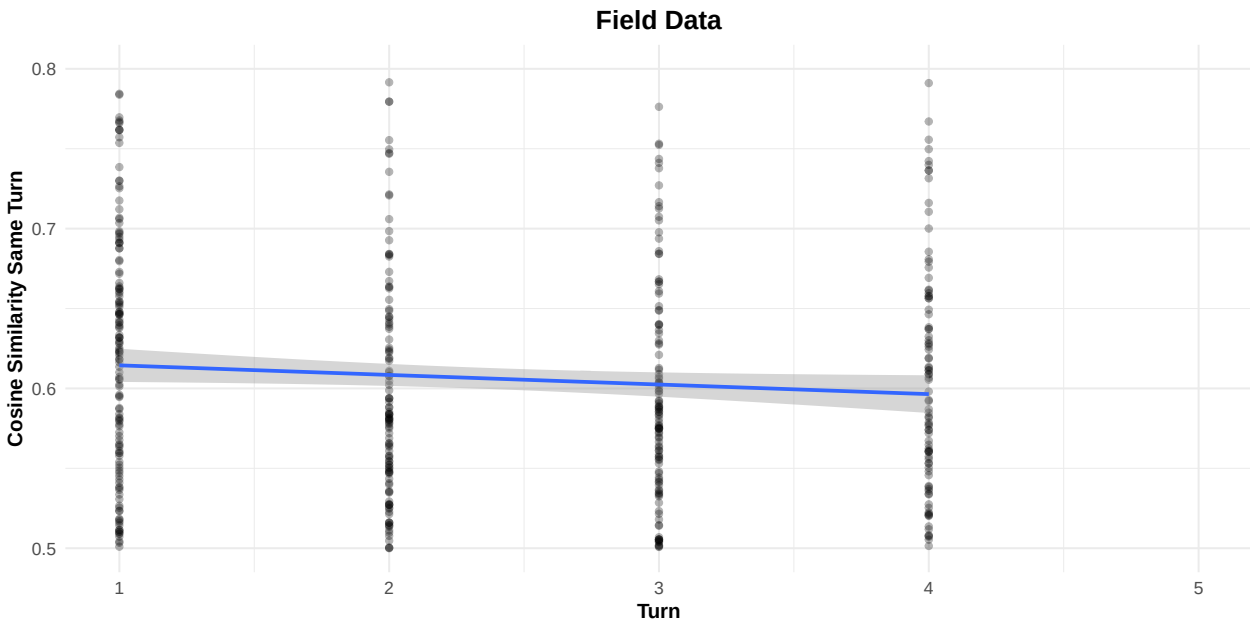


4.2 4.2 Cosine Similarity Mixed Model

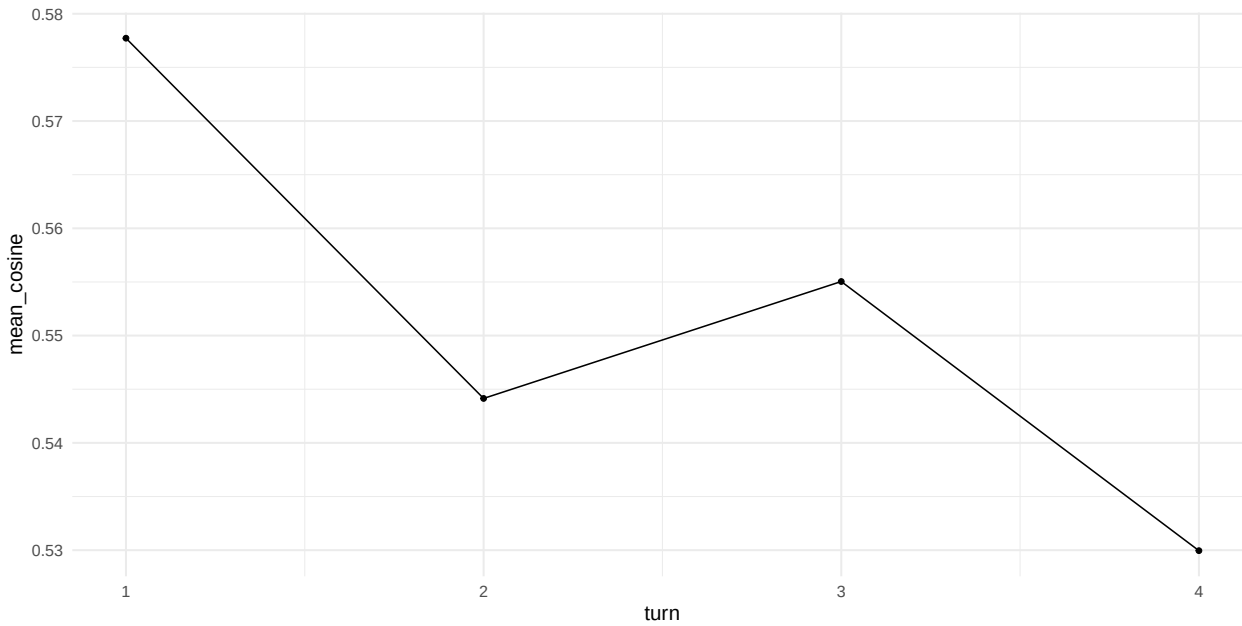
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: cosine_sim_same ~ turn + (1 | conversation_id)
##    Data: df_cosine_sim
##
## REML criterion at convergence: -1110.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.40623 -0.70494 -0.02459  0.61946  3.07832
##
## Random effects:
##  Groups          Name        Variance Std.Dev.
##  conversation_id (Intercept) 0.001212 0.03482
##  Residual                    0.011234 0.10599
## Number of obs: 722, groups:  conversation_id, 205
##
## Fixed effects:
##               Estimate Std. Error         df t value Pr(>|t|)
## (Intercept)   0.585518   0.009663 714.164260  60.594  < 2e-16 ***
```

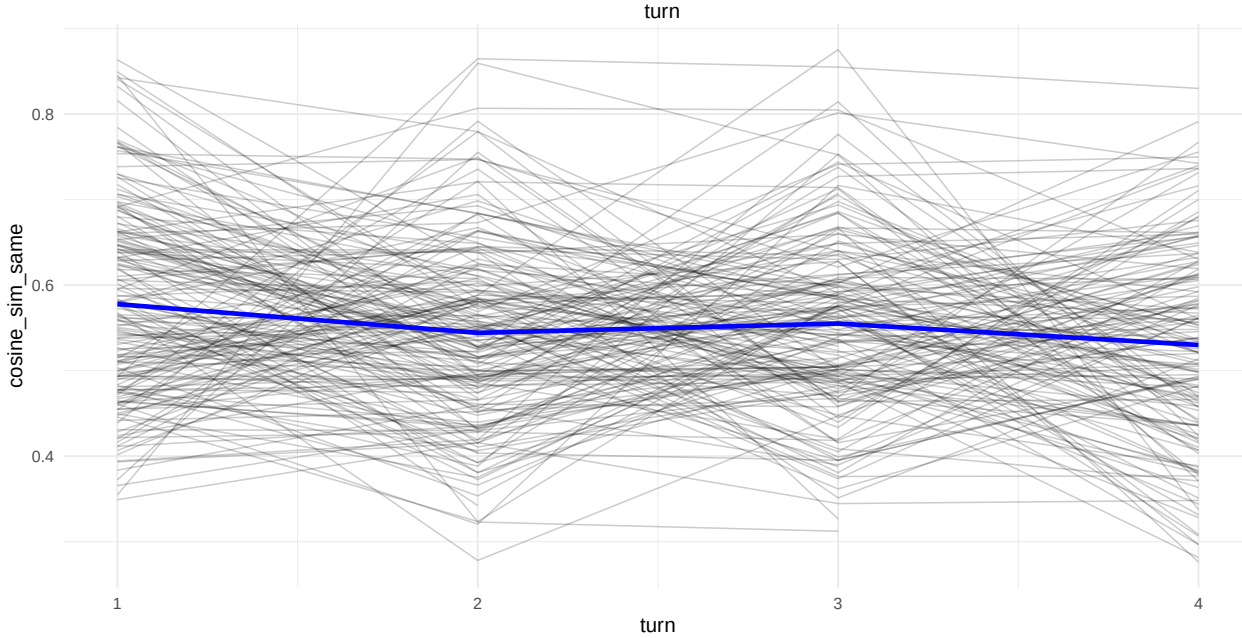
```
## turn         -0.013583   0.003558 568.465513  -3.818 0.000149 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr)
## turn -0.875
```

4.3 4.3 Cosine Similarity Linear Plot



4.4 4.4 Cosine Similarity Over Time





4.5 4.5 Cosine rubber band

```
## === Turn-Level Rubber-Band Cosine Distance: Field Data ===

##
## Call:
## lm(formula = dist_change_next ~ dist_change, data = rb_cosine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.41500 -0.09909  0.00418  0.10399  0.49463
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.056659   0.006391   8.865   <2e-16 ***
## dist_change -0.562391   0.045070 -12.478   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.138 on 465 degrees of freedom
## Multiple R-squared:  0.2509, Adjusted R-squared:  0.2492
## F-statistic: 155.7 on 1 and 465 DF,  p-value: < 2.2e-16
```

4.6 4.6 Turn level Cosine Permutation test

```
## === Cosine Distance Turn-Level Permutation Test Results ===
```

```
## Observed slope: -0.5624
```

```
## Null mean: -0.0451
```

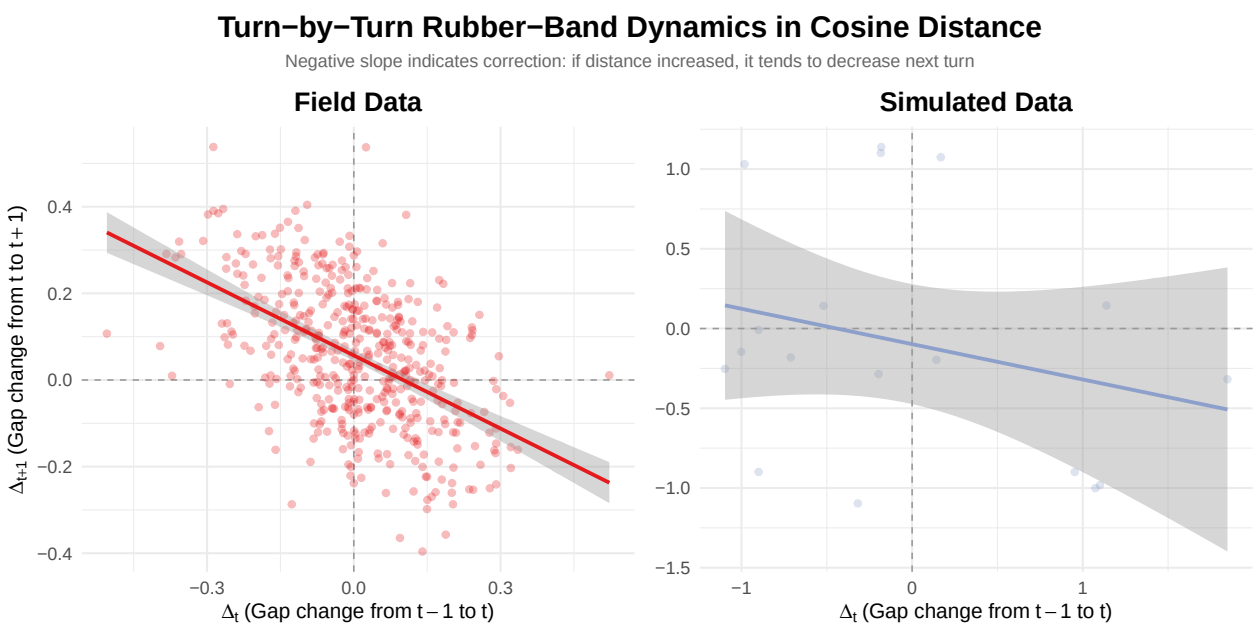
```
## Null SD: 0.069
```

```
## Effect size (z): -7.5
```

```
## P-value: <2e-16
```

```
## # A tibble: 2 x 4
##   transition_type rb_slope correlation     n
##   <chr>              <dbl>       <dbl> <int>
## 1 Stage Boundary    -0.472      -0.511   274
## 2 Within-Stage      -0.701      -0.627    19
```

4.7 4.7 Turn Level Cosine RB Linear plot

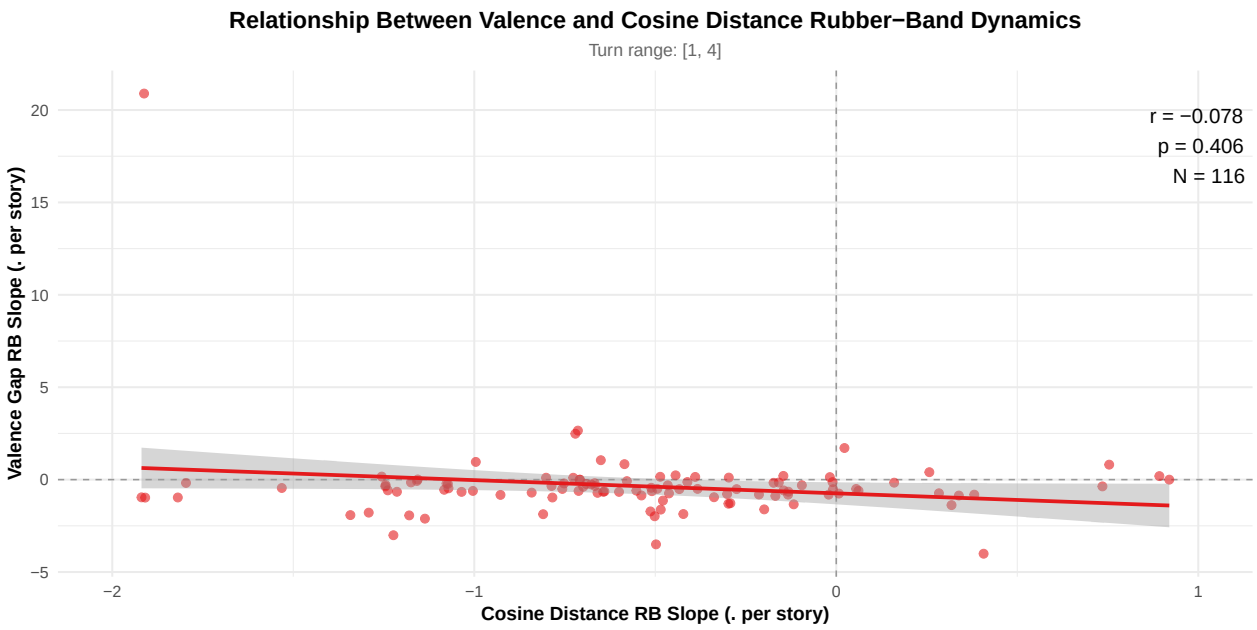


4.8 4.8 Correlation Between Cosine Distance and Valence Rubber-Band

```
## === Correlation Between Valence and Cosine Distance Rubber-Band ===
```

```
## Turn range: [1, 4]
```

```
## N observations (turn-level): 448
## Turn-level correlation of gap changes (Δt): -0.066
##   Interpretation: Do valence and cosine gaps change in the same direction
##                   at the same turn across all stories?
## Turn-level correlation of next changes (Δt+1): 0.05
##   Interpretation: Do subsequent valence and cosine corrections correlate?
## N stories with >= 3 turns: 116
## N stories with >= 3 turns: 116
## STORY-LEVEL correlation of RB slopes: -0.078
##   Interpretation: Across stories, does valence RB strength predict cosine RB strength?
##                   Positive r means stories with strong valence correction
##                   also show strong cosine distance correction.
##   95% CI: [-0.256, 0.106]
##   p-value: 0.406
##   df: 114
## Shared variance (R²): 0.006
## Interpretation: 0.6 % of variation in valence RB slope
##                 is explained by cosine distance RB slope
```




```
##
## === Regression: Cosine RB Predicting Valence RB ===
##
## Call:
## lm(formula = valence_rb_slope ~ cosine_rb_slope, data = rb_slopes)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -3.5223 -0.4117 -0.1597  0.2484 21.0977
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)      -0.4379     0.2280  -1.921   0.0573 .
## cosine_rb_slope  -0.1208     0.1449  -0.833   0.4064
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.183 on 114 degrees of freedom
## Multiple R-squared:  0.006055,   Adjusted R-squared:  -0.002663
## F-statistic: 0.6945 on 1 and 114 DF,  p-value: 0.4064

##
## === VALIDATION CHECK ===

## Correlation from cor(): -0.078

## Correlation from lm R²: 0.078

## These should match (they do if correlation is positive).
```